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System and method for expediting the delivery of food orders to customers

Abstract

Automated food delivery methods and systems using computer-managed delivery to an array of food delivery enclosures and computer-enabled customer access to the food delivered to the enclosures. Customers using the methods and systems may utilize their personal communication device such as a cell phone or tablet to order food and subsequently open an enclosure containing their food order.

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Background/Summary

CROSS REFERENCE OF RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 18/057,043 filed Nov. 18, 2022, which is a divisional of U.S. application Ser. No. 17/308,844 filed May 5, 2021, now U.S. Pat. No. 11,514,742, which claims the benefit under 35 U.S.C. 119(e) of U.S. Provisional Application No. 63/101,627, filed May 7, 2020, and U.S. Provisional Application No. 63/205,483, filed Dec. 15, 2020, the entire contents of which are incorporated herein as if fully set forth herein.

FIELD OF INVENTION
(1) This invention relates generally to the delivery of food orders to customers using an array of enclosures in which food orders are placed by a supplier and from which food orders are retrieved by customers.
TRADEMARK AND COPYRIGHT RESERVATION
(2) A portion of the disclosure contains material which is subject to trademark and copyright protection. The trademark and copyright owner(s) has (have) no objection to reproduction of the disclosure in the form in which it appears in the files of the US patent & Trademark Office, but otherwise all rights are reserved. Copyright ?2019 Automat Kitchen, LLC.
BACKGROUND OF THE INVENTION
(3) U.S. Pat. Nos. 7,568,618 and 8,261,980, which are incorporated herein by reference, describe a system and method for delivering food to customers using an array of enclosures. Each of the enclosures includes a food supply opening through which food is placed in the enclosure and a food delivery opening through which food is delivered to the customer after a door which covers the delivery opening is unlocked and opened by the customer. A similar system and method are described in U.S. patent application Ser. No. 15/247,511 filed Aug. 25, 2016, now U.S. Pat. No. 10,482,525.
SUMMARY OF INVENTION
(4) The patent claims summarize many of the important aspects of the invention, and these and other important aspects are described in the following specification as well as the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The detailed description of the preferred embodiments and various important aspects of the invention will be better understood with reference to the following drawings:
(2) FIG. **1** illustrates a food ordering and delivery system employing an array of food enclosures to deliver food to customers and FIG. **1**A illustrates a kiosk for ordering food.
(3) FIG. **2** is a plan view of a restaurant employing the food ordering and delivery system illustrated in FIG. **1** and a plurality of the kiosks illustrated in FIG. **1**A;
(4) FIG. **3** is a schematic representation of the food ordering and delivery system illustrated in FIGS. **1**, **1**A and **2**.
(5) FIG. **3**A is a flow chart for the system shown in FIG. **3**;
(6) FIG. **4**A is a frontal perspective view of one of the enclosures in the array shown in FIG. **1** in a closed condition.
(7) FIG. **4**B is a rear perspective view of the enclosure shown in FIG. **4**A shown from the back of the enclosure which has been opened in the rear to permit food to be supplied to the enclosure.
(8) FIG. **4**C is a frontal perspective view of the enclosure of FIG. **4**A in a partially open position.
(9) FIG. **4**D is a frontal perspective view of the enclosure of FIG. **4**A which is fully open in the front

to permit food to be delivered or retrieved from the enclosure to or by a customer.
(10) FIG. **5**A is a frontal perspective view of the enclosure of FIG. **4**A in the closed position displaying a promotional image on a screen.
(11) FIG. **5**B is a frontal perspective view of the enclosure of FIG. **4**A displaying an enclosure number and a customer name.
(12) FIG. **6** is a view of a screen at the top of the array shown in FIG. **1** for displaying customer orders which are ready for delivery.
(13) FIG. **7** is a view of a screen at the top of the array shown in FIG. **1** for displaying customer orders which are in the process of being prepared.
(14) FIG. **8**A is a view of screens on two adjacent enclosures in the array shown in FIG. **1** including the screen shown in FIG. **5**B.
(15) FIG. **8**B is a view of the two adjacent screens of FIG. **8**A after customer interaction with the screens shown in FIG. **8**A.
(16) FIG. **8**C is a view of the two adjacent screens of FIG. **8**B after customer interaction with the screen shown in FIG. **8**B.
(17) FIG. **8**D is a view of the screen shown in FIGS. **8**A and **8**B after the customer has removed his/her order from the enclosure.
(18) FIG. **9** is a view of a screen on an order preparation monitor in a food supply area behind the array of enclosures showing a queue of orders to be prepared.
(19) FIG. **10** is a view of a screen on an order control monitor in the food supply area behind the array of enclosures to enable the assignment and delivery of different food orders to the enclosures in the array.
(20) FIG. **10**A is a view of a pop-up on the screen of FIG. **10** which confirms the delivery of an order to specific enclosures.
(21) FIG. **10**B is a view of another pop-up on the screen of FIG. **10** which confirms the status of an order in a particular enclosure.
(22) FIG. **11** is a diagram of an algorithm for generating, sequencing and preparing customer orders.
(23) FIG. **12** is a diagram of an algorithm for assigning and supplying customer orders to various enclosures in the array.
(24) FIG. **13** is a diagram of an algorithm for delivering a food order from an enclosure in the array.
(25) FIG. **14** is a chart depicting an algorithm for and the various steps in a method by which a customer orders food to be delivered to an enclosure by using a computer, tablet, smart phone or other portable personal device.
(26) FIG. **15**A illustrates four sequentially presented images on the screen of an enclosure receiving a food order which the customer can retrieve using their portable personal device without touching the enclosure surfaces.
(27) FIG. **15**B is a view of four sequentially presented images presented on the customer's portable personal device for opening the enclosure behind the screen shown in FIG. **15**A without touching the screen or any other enclosure surface.
(28) FIG. **16** is a chart depicting an algorithm for and the various steps in a method by which a customer orders food to be delivered to an enclosure by using a kiosk in the restaurant.
(29) FIG. **17** is a view of four sequentially presented screens for opening an enclosure using the touch screen of the enclosure.
(30) FIG. **18**A illustrates an alternative configuration for the display on the screen of the order control monitor in the food supply area showing no orders in the order queue and all 20 of the enclosures available.
(31) FIG. **18**B illustrates a display on the screen of the order control monitor at a later point in time showing 6 orders in the order queue and all 20 enclosures still available.
(32) FIG. **18**C illustrates the display of FIG. **18**A at a slightly later point in time than that illustrated in FIG. **18**B showing 1 of the 6 orders in the queue being loaded in 2 of the 20 available boxes.
(33) FIG. **18**D illustrates the display of FIG. **18**A at a point in time when there are 4 orders in the

queue, 5 enclosures holding orders and 1 enclosure disabled.
(34) FIG. **18**E illustrates the display of FIG. **18**A at a later point in time after 1 order has been loaded into 2 adjacent boxes leaving 18 available enclosures for the 1 order in the queue.
(35) FIG. **18**F illustrates the display of FIG. **18**A with 6 orders in the queue and 9 enclosures available with 11 enclosures unavailable for various reasons.
(36) FIG. **19** illustrates a display on a monitor advising customers of the status of orders received but not yet in the order queue.
(37) FIG. **20** illustrates a display on a monitor advising customers of orders which have been prepared and placed in enclosures.
(38) FIG. **21** illustrates the display on the monitor shown in FIG. **20** advising customers of orders using all 20 enclosures.
(39) FIG. **22** illustrates a display on the monitor advising customers of orders prepared and placed in the enclosures which also alerts a single customer for a brief period of time that their order has just been placed in an enclosure.
(40) FIG. **23** is a flow diagram illustrating the steps in an algorithm for generating the various displays on the order control monitor as shown in FIGS. **18**A-F.

DESCRIPTION OF PREFERRED EMBODIMENTS

(41) In FIG. **1**, a food ordering and delivery system embodying the invention disclosed including food ordering kiosks **12** and a wall **14** incorporating an array **16** of food enclosures or boxes **18**. Each of the enclosures **18** includes a touch screen door **19** which will automatically open as will be described in greater detail with respect FIGS. **4**A-D so as to permit food to be delivered to a customer **20** standing in front of the array **16**. The wall **14** includes promotional/informational screens **22** and **24** located at the top of the wall **14** above the enclosures **18**. In addition, the wall **14** includes a customer order “READY” monitor **26** for displaying a customer name and enclosure number from which a food order can be delivered to the customer as will be described in greater detail with reference to FIG. **6**. The wall **14** also includes a customer order status or “IN PROCESS” screen **28** for displaying the status of a customer order as will be described in further detail with reference to FIG. **7**. The kiosks **12** include a touch screen **30** as shown in FIG. **1**A for ordering specific items which appear on the touch screen **30** as well as a slot **32** for receiving a credit or debit card for charging the cost of the food order as also shown in FIG. **1**A.
(42) The food ordering and delivery system shown FIGS. **1** and **1**A is incorporated into a food service facility such as a restaurant as shown in FIG. **2** wherein the wall **14** separates a customer accessible area **34** including a seating area **36** with tables and chairs **38** as well as the kiosks **12** and a food preparation and supply area **38** which includes kitchen equipment **40**, counters **42** and storage areas **44**. In accordance with one important aspect of the preferred embodiments, the food preparation area **38** includes one or more preparation monitors **46** visually and physically accessible to kitchen personnel who prepare food orders. These monitors **46** display all customer orders as will be described in greater detail with reference to FIG. **9**. In further accordance with this important aspect, the food preparation area **38** also includes at least one kitchen expediter or control monitor **48** visually and physically accessible to kitchen order expediters directing and supplying prepared orders to available enclosures in the array **16** as will be described in greater detail with reference to FIG. **10**.
(43) As shown in FIG. **3**, the wall **14** and the kiosks **12** are under the control of a computer system comprising digital processing hardware and a digital memory for storing software including programmed instructions. The digital processing hardware includes a master computer or “Wall Computer” **44** and an ordering and billing computer or “Order Computer” **45**, and the digital memory includes a data storage unit **47** in hardwired (as shown) or wireless communication with the master computer **44** and the ordering and billing computer **45**. The ordering and billing computer **45** which may be on-site as shown or off-site is hardwired or in wireless communication with a server (not shown) hosting a website of the restaurant and/or a restaurant APP which customers may access using various portable, personal devices such as cell phones, tablets and lap tops. The ordering and billing computer **45** is also in communication with the personal devices of customers who are on-line with the

restaurant website or the downloaded restaurant APP which facilitates the ordering of food. The point of sale (POS) functions provided by the ordering and billing computer **46** are commercially available using APIs such as those provided by Toast, Inc. and Panasonic. After a customer has entered and paid for his or her order at a kiosk or on a personal device using the restaurant APP or website, the customer will receive a customer specific code, preferably an at least three-digit numerical code, which will allow them to open one or more food enclosures **18** as will be described below.

(44) FIG. **3** also depicts the various screens on the wall **12** including touch screens incorporated into each of food delivery doors **19** on the enclosures **18** which are coupled to and under the direct control of the master computer **44** or indirect control of the master computer if an individual microprocessor is provided for each touch screen doors **19**. The customer order alert or READY ORDER screen **26** as well as the customer order status or IN PROCESS screen **28** are also coupled to and under the direct or indirect control of the master computer **44** depending on whether or not individual microprocessors are provided for these screens. In the same manner, the promotional/informational screens **22** and **24** are coupled to and directly or indirectly controlled by the master computer **44**.

(45) FIG. **3** also depicts the nature of the displays on the various touch screens in the doors **19** of the enclosures **18**. When an enclosure does not contain a food order, a promotional and/or decorative display such as that shown in FIG. **5**A will appear on the touch screen door of the enclosure. As shown in FIG. **5**A, the display is static but it may also be dynamic; i.e., a video. When an enclosure does contain a food order, the number of the enclosure **18** and name of the customer appears on the touch screen door **19** as also shown in FIG. **5**B. In accordance with one important aspect, an order may require and therefore be placed in a plurality of adjacent enclosures. Under these circumstances, peripheral portions **62** of the screens on two adjacent enclosures are highlighted as shown in FIG. **3** and FIGS. **8** (A-D).

(46) The flow chart of FIG. **3**A describes the various steps of customer ordering process and the delivery of food orders using the system of FIG. **3**. The customer initially decides in Step **120** whether to go to the restaurant or other food service facility to order (Step **122**) or to use a web site or App of the restaurant or food service facility for ordering (Step **120**). If the customer goes to the restaurant or food service facility to order, the customer will order at one of the kiosks **12** and use the kiosk touch screen **30** to provide the necessary order information (Step **124**). If the customer uses the web site or App to order, the order is remotely entered (Step **126**). Once the order is entered regardless of the location of entry, the order is transmitted to the food preparation area **38** under the control of the order and billing computer **45** for preparation and deliver to a box **18** (Step **130**). When the order is ready, a kitchen expediter assigns a box(es) **18** for receiving the delivery to the customer and the customer is notified that their order is ready by sending a message to the customer's personal device (Step **132**). The customer's food order is then placed in a box **18** where the customer may collect it after entering an assigned code whereupon the door **19** of the box **18** opens (Step **134**). After the customer collects the food order, the door **19** of the box **18** automatically closes (Step **136**).

(47) In FIG. **4**A, a single enclosure **18** of the wall **14** is shown in the closed condition with the touch screen door **19** closing the food delivery opening at the front of the enclosure **18**; i.e., on the side facing the customer accessible area **34** as shown in FIG. **2**. The touch screen door **19** and sides **64** form the enclosure along with a hinged door **66** on the food preparation side **38** which is shown in the open position in FIG. **4**B but can be closed and maintained in the closed position by magnetic catches **68**. When the hinged door **66** is in the open position as shown FIG. **4**B, a food order may be supplied through an opening **72** before being closed prior to initiating the opening of the touch screen door **19** as shown in FIG. **4**C and before the touch screen door **19** is fully raised up and retracted like a garage door into the enclosure **18** as shown in FIG. **4**D by a motor driven retraction mechanism. When the touch screen door **19** is in the closed position before a food order **70** has been delivered to the enclosure through a food supply opening **72** at the rear of the enclosure **18**, the touch screen door **19** displays a promotional and/or decorative image or video as shown in FIG. **5**A. After the food order has been supplied to the enclosure **18**, the number of the enclosure **18** and the name of the customer for the order is displayed on the touch screen door **19** as shown in FIG. **5**B; i.e., “#2 John H”.

However, as shown in FIGS. **3** and **8**A, the "John H." order, because of its size has necessarily been delivered to two enclosures; i.e., "#2" and "#3" and peripheral portions **62** of the touch screen doors **19** on those enclosures are illuminated to assure that the customer "John H." understands that his order is contained in two adjacent enclosures **18**. When a customer such as "John H." is ready to open enclosures "#2" and "#3", he taps on the touch screen door **18** as shown in FIG. **8**A and new display appears on the touch screen **19** of enclosure "#2" as shown in FIG. **8**B.

(48) In accordance with one important aspect of the invention, the touching of the touch screen in FIG. **8**A enables the display shown in FIG. **8**B which represents a key pad **74** for entering an assigned code and three windows **76** for displaying the code entered using the key pad **74**. As shown, only the code numbers "2" and "3" have been entered and are displayed in the first and second window. The customer will have received the assigned code which may be displayed on the touch screen **30** of a kiosk **12** or printed at the kiosk **12**. If the customer used his own personal device to make his/her order by using an application downloaded to his device or accessing the restaurant website, he may receive his assigned code by a text or e-mail delivered to that device or by other means enabled by using the restaurant APP or website.

(49) In accordance with another important aspect, entry of the code using the key pad will produce another new display on the touch screen **19** as shown in FIG. **8**C if the code which the customer entered was incorrect. By tapping on a button indicting that the customer can try "again", the display of key pad **74** as shown in FIG. **8**B will be presented again. If the customer decides to abandon his efforts, he can tap on a "cancel" button.

(50) If the customer enters the correct code using the key pad **74**, the touch screen door **19** will open. In accordance with another important aspect, the touch screen door **19** will remain open a predetermined period of time so as to allow the customer to remove his food order from the enclosure **18**. After the predetermined period expires and the touch screen door **19** has closed as shown in FIG. **8**D, the touch screen door will present a new display asking the customer whether he is "all done" which provides him with an opportunity to confirm by tapping on an "all done" button. If he is not done, he can so indicate by tapping on an "open again" button and the door **18** will again open for the predetermined period time. Preferably, the predetermined period of time is more than 15 and less than 60 seconds with approximately 30 seconds considered most preferred so as to provide sufficient time for the customer to remove the order but also return the enclosure to an available status as soon as possible to efficiently handle any order backlog created by other customers.

(51) The display on the screen of the kitchen order preparation monitors **46** as shown in FIG. **9** includes customer names and the items in the order which need to be prepared. The orders are presented from top to bottom of the screen in the sequence that orders are to be prepared; i.e., those higher on the screen are prepared sooner. In accordance with another important aspect, the monitors **46** not only advise the kitchen personnel what to prepare in order to fulfill orders, the monitors **46** under the control of the billing and ordering computer **46** creates a queue of orders to be prepared based on the chronological receipt of orders from the kiosks **12**, or for orders placed on personal devices, the specified pick-up time less a predetermined period of order preparation time. In other words, if a customer specifies a pick-up time of 1:30 PM on his or her personal devices, the order is placed in the queue as if it had been ordered at a kiosk **12** at 1:25 PM assuming that the predetermined period for the order preparation time is 5 minutes.

(52) The display on the screen of the kitchen expediter monitor **48** includes representations **78** of each of the enclosures **18** in the array **16** as shown in FIGS. **1** and **3** and representations of orders **80**; e.g., orders from "John H." and "Bob J". Order expediters in the kitchen area are able to drag the representation of an order **80** to a representation **78** of one or more enclosures. For example, the representation **80** of the order of "John H." can be dragged to representations **78** of enclosures #2 and #3. The expediter may then open the hinged door **66** on enclosures #2 and #3 and place the "John H." order in those enclosures. By tapping or clicking on the representation of enclosure #2 on the screen of the expediter monitor once, a pop-up display as shown FIG. **10**A appears showing the enclosures involved in his order. The expediter can then confirm that the order has been delivered to the

enclosures specified in the pop-up by tapping or clicking on the “OK” button or cancel by tapping on the “CANCEL” button. The expediter can then tap or click on the representation **78** twice to signify that the order is complete and then tap or click on the “ANNOUNCE” button and the customer's name and enclosure number will appear on the screen of the order ready monitor **26**.

(53) In accordance with another important aspect, the expediter is alerted to the status of each of the enclosures through the use of color coding to signify the availability of each enclosure **18** in the array. For example, the representation **78** of the enclosure may be colored blue if the corresponding enclosure is available for a new delivery. The representation **78** may be colored green if an order has recently been placed in the corresponding enclosure. The representation **78** may be colored yellow if the touch screen door of the corresponding enclosure has been opened and closed after the timing out of the predetermined time for automatic closing (e.g., 30 seconds). The representation **78** may also be colored red if a prolonged period of time has passed since an order was placed in the corresponding enclosure. The expediter can also tap or click once on any representation and a pop-up display will appear as shown in FIG. **10**B. The pop-up will not only identify the customer and order code but will also display how long the order has been in the corresponding enclosure expressed in minutes and seconds. If the expediter determines that an order should not be supplied to any enclosure or be removed from any enclosure because of the length of time that an order has been in the enclosure, he may physically remove the order from the enclosure and also drag the order out of the corresponding representation **78**.

(54) The specific steps of an algorithm used by the orders computer **45** for performing Steps **120** through **130** of the flow chart of FIG. **3**A for the orders computer **45** are shown and described in FIG. **11**. More specifically, Steps **140**, **142**, **144** and **146** (kiosk ordering), Steps **148**, **150**, **152**, **154** and **156** (remote device ordering), and Steps **158**, **160** and **162** (order preparation information) as described in FIG. **11** are all performed by and under the control of the orders computer **45**.

(55) The specific steps of an algorithm for performing Step **132** of the flow chart of FIG. **3**A are shown and described in FIG. **12**. More specifically, Steps **168**, **170**, **172**, **174**, **176** and **178**, as described in FIG. **12**, all relate to displays on and used of the expediter monitor **48** as enabled and controlled by the wall computer **44**. Steps **164** and **166** are performed by and under the control of the orders computer **45**.

(56) The specific steps of an algorithm used by the wall computer **44** for performing Steps **134** and **136** of the flow chart of FIG. **3**A and providing the functionality described with respect to FIGS. **8**(A-D), **9**, **10**, **!0**A and **10**B are shown and described in FIG. **13**. More specifically, Steps **180**, **182**, **184**, **186**, **188**, **190**, **192**, **194**, **196** and **198** all related to the displays and interaction of the customer with the displays on a touch screen **19** of a door **18** of a box **14** before the door **18** opens. Steps **200**, **202**, **204**, **206**, **208** and **210** all relate to the displays on the door screen **19** after the door **18** opens.

(57) As described above, the customers touch the touch screen doors **19** to enter the customer specific codes to open the doors on the enclosures to access their orders. The customers may subsequently need to touch the doors **19** if they enter incorrect codes or need to open the doors again. This touching potentially exposes the customers to germs including extremely contagious viruses. Although the customers can use their personal devices to order their food and avoid the use and touching of the touch screens **30** on the kiosks **12**, images on the touch screen doors **19** as shown in FIGS. **8**(A-D) may be transmitted over the Internet from the Wall Computer **50** to the website server **58** and the server for the Device APPs **60** as shown in FIG. **3** so as to allow the customers to interact with the displays of FIGS. **8** (A-D) on their personal devices without ever touching the touch screen doors **19**. Another embodiment of the invention in which personal devices may be used to order food and open the doors **19** on the enclosures in a touchless manner will now be described with reference to FIGS. **14** and **15**A&B.

(58) As described in FIG. **14**, a customer orders using their personal device without using or touching any of the kiosks **12**. In Step **1**, the customer accesses the restaurant website using the browser of their personal device to obtain restaurant information including the menu. In the alternative, the customer can access the information available on the website by using an APP stored on their personal device.

By touching an “Order Button”, the customer is placed in communication with the order and billing computer **46** which may be on-site at the restaurant as shown in FIG. **3** or off-site and provided by a third-party point-of-sale (POS) service entity such as Toast using the WEB. With this communication, the customer is able to perform Step **2** including the review and selection of menu items which are displayed as one or more images on the screen of the personal device. This is followed by Step **3** wherein the customer pays for the order, specifies a pick-up time and identifies a contact e-mail address and telephone number. In Step **4**, the POS entity then provides an order confirmation by e-mail. Using SMS, the restaurant provides the customer with advisories indicating that their order is “in progress” in Step **5** and that the order is “almost ready” in Step **6**. In Step **7**, the customer is provided with an SMS text providing a customer specific, pick-up code and a customer-specific URL to be used for a touchless pick-up. Although a customer order number provided by the POS system could be used as the pick-up code, the pick-up code is preferably randomly generated in this embodiment.
(59) When the customer is ready to pick-up their order, they will be presented with a series of displays on the touch screen door **19** of the enclosure receiving their order as shown in FIG. **15**A. Display **80** appears on the screen before the customer's order is placed in the enclosure **18** behind the touch screen. After the order for a particular customer (e.g., “Claire”) has been placed in the enclosure number “16”, the display **82** appears on the touch screen door **19** so as to advise and/or confirm that the order of the customer “Claire” has been placed in enclosure number “16”. If the customer “Claire” does not choose to use a touchless opening of the enclosure, she can touch the display **82** and will then be presented with display **84** confirming her order number “259” and presenting a keypad which she can use to open the door of the enclosure by entering her pick-up code which is randomly generated and different from her order number. If the door will not open after the pick-up code is entered or any other problem or issue arises, “Claire” can touch the word “HELP” on the display **84** and the display **86** will appear. “Claire’ then is presented with various problems/issues which she can tap in an effort to provide a solution to “Claire's” problems/issues.
(60) If a customer such as “Claire” confirms that she wants to open the door of the enclosure **16** in a touchless manner using the customer-specific URL received by e-mail, the displays of FIG. **15**B will serially appear on the personal device of the customer. A display **88** confirms that a touchless pickup for enclosure (“BOX”) **16** is enabled. By touching or clicking on that screen showing a button defined by a circle containing a knife and fork, image **90** will appear showing a keypad and instructions to enter the randomly generated pick-up code which is displayed on the touch screen of the enclosure (“Box”)**16** and was received by SMS in step **7** of FIG. **14**. If the customer correctly enters the pick-up code, image **92** will appear confirming on the door of the “BOX” **16** as well as the “BOX” **17** containing the customer's order. Image **94** appears if no pick-up code is displayed on the assigned enclosure (“BOX”) with instructions to call a staff member. Note that help can also be summoned by touching or clicking on the word “HELP” in the displays **88**, **90**, and **92** and prior displays may also be accessed using the “BACK” buttons.
(61) As an alternative to entering the pick-up code on a keypad appearing on the customer's personal device for touchless opening, touchless opening may be achieved by the customer by using the SMS in the following way. The customer first goes to the SMS text provided in Step **7** of FIG. **14** which contains the pick-up code. The customer then uses the text to open the door in one of three different ways: (1) the customer can click on the pick-up code in the received text; (2) the customer can send a reply text containing the pick-up code; or (3) the customer can send a reply text consisting of the word “open”.
(62) Another alternative for touchless opening involves the use of a QR code. A customer-specific QR code is contained in an SMS message received by the personal device of the customer. The wall **14** is capable of reading the QR code on the screen of the personal device and then transmitting a customer specific URL to that device which is unique to the customer. The customer can then open the door and perform other functions such as seeking help by interacting with the display on the screen(s) provided by the URL. This assures that the door will only open when the customer is physically present in the restaurant to scan in their QR code. In addition, the URL may be used to access the displays like those

shown in FIG. **15** so as to allow the customer to enter a pick-up code delivered by SMS on a keypad so as to provide a double layer of security.

(63) Although touchless ordering and door opening is available, the customer may still choose to order at a kiosk by touching the kiosk screen and opening the door on a food enclosure by touching the screen on the door. A series of ordering steps are shown in FIG. **16** which are performed using a touch screen at a kiosk. In Step **1**, the customer is presented with a screen saver. When the customer touches the screen saver, an image is displayed which provides the customer, in Step **2**, with the ability to create an order by reviewing a menu and selecting menu items. In Step **3**, the customer is presented with one or more screen images permitting the customer to place an order, pay, provide a pick-up time, provide a contact phone, and then, at the customer's option, enable a print-out of a receipt which includes the pick-up code. In Step **4**, the image presented at the kiosk includes the pick-up code which is also delivered to the customer's contact phone by SMS.

(64) When the customer's order is ready for pick-up from an enclosure, the serial images on touch screen of the enclosure containing the order as shown in FIG. **17** are substantially identical to those on the touch screen of the enclosure shown in FIG. **15**A. However, if a customer order number is used as the customer-specific code and not a randomly generated code, display **84**A as shown in FIG. **17** will not, for security reasons, include the customer order.

(65) As in the embodiment of the invention shown in FIGS. **8**A-D, the doors of the enclosures (or “BOXES”) **18** in the embodiment of the invention described using FIGS. **14-17** may be set to close automatically a predetermined period of time after opening. Since a customer may not be able to remove all of the items in one or more assigned enclosures during this holding period, customers need to be provided with an opportunity to reopen the enclosures. Accordingly, a display will appear on the touch screen of the enclosure door, as well as on the personal device of a customer operating in the touchless mode, which can used to reopen the door for another predetermined period time.

(66) The screens of the order control or expediter monitor **48** in this embodiment of the invention will now be described in detail with reference to FIGS. **18**A-F. As in the earlier described embodiment as shown in the displays shown in FIG. **9**, the monitor **48** is under the control of the “Wall Computer” **44** which, in this embodiment, operates under stored program instructions so as to allow the expediter to interact with the screen by clicking or tapping on the screen to perform various functions and view various displays on the screens which may be color coded to indicate various conditions as will now be described.

(67) In FIG. **18**A, all 20 enclosures in the wall **14** are depicted in the display by numbered squares **100** with enclosure status displays **102** immediately below the numbered squares. As shown, the status displayed for each of the displays **102** indicates that the respective enclosures or “BOXES” are available to receive an order (e.g., “Available”) consistent with a status summary displayed in the area **104** indicating all “20 Boxes Available”. An area **106** above the area **104** is available to display an order queue which is empty in FIG. **18**A which is consistent with an order summary displayed in area **108** indicating “No orders in queue”.

(68) The display on the screen as shown in FIG. **18**B still shows all “20 Boxes Available” but an order queue is now displayed in the area **106**. The order queue which is chronological by specified pick-up times provides four pieces of information: (1) a customer name (e.g., “Katie”); (2) a customer specific order number (e.g., “#222” for “Katie”); (3) a time that the order was left (e.g., “9:56” for “Katie”); and (4) a time of pick-up specified (e.g., “10:00” for “Katie”). An area immediately below each customer name displays an “ASSIGN” button **109** on which the expediter can click or tap to initiate an assignment of an order to an enclosure or “Box”.

(69) FIG. **18**C illustrates the enclosure or “Box” assignment process using the order of “Katie” as an example. As shown, there are still 6 orders in the queue but the “ASSIGN” button **109** shown in FIG. **18**B under “Katie” has been clicked or tapped on to initiate the assignment for the order of “Katie” to enclosures or “Boxes” **1** and **2**. Because of the size of the “Katie” order, the expediter has determined that two enclosures or “Boxes” will be required and the expediter has therefore clicked or tapped on two adjacent boxes for the convenience of “Katie”. Of course, a smaller order would only require the

selection of one enclosure or “Box”. After the expediter has loaded the “Katie” order into the enclosures or “Boxes” **1** and **2**, the expediter can confirm that the order has been loaded into the enclosures by clicking or tapping on a “CONFIRM” button **110**. This will remove the “Katie” order from the order queue in the display of FIGS. **18**B&C and show the “Katie” order in enclosures or “Boxes” **1** and **2** in the display of FIG. **18**E which shows only “Chad” remaining in the order queue and then only 18 boxes available with the “Katie” order shown in the area **102**. In the alternative, the expediter can cancel the enclosure assignment by clicking or tapping on a cancel or “X” button **112** shown in FIG. **18**C and the “Katie” order will return or remain in the order queue as shown in FIG. **18**B.

(70) FIGS. **18** D-F show the status of all enclosures or “Boxes” which are not designated as “available” so as to provide the expediter with important information in the area **102**. For example, the areas **102** for enclosures or “Boxes” **1** and **2** shown in FIG. **18**E display information which indicates the order number (i.e., #222) for the “Katie” order has been in the enclosures for 5 minutes after the specified pick-up time. In FIG. **18**D, the area **102** for enclosures or “Boxes” **8** and **17** display, by way of example, information for the “Maria” order #125 and the Janet order #102. The minus sign indicates that the specified pick-up time for the “Maria” order will not occur for another minute (i.e., the order was necessarily placed in the enclosure prior to the specified pick-up time). In the case of the “Janet” order, 25 minutes have elapsed beyond the specified pick-up time which exceeds a predetermined limit and that fact is denoted by displaying the word “Expired”. In order to be sure that the expediter is aware of this expiration and can then remove the order from the enclosure, the words and numbers may be color coded red. Red color coding may also be used to indicate the number of minutes that have elapsed since the specified pick-up time even though the predetermined limit has not been reached as in the case of the “Katie” order. Green color coding may be used for the number of minutes prior to the specified pick-up time as in the case of the “Maria” order.

(71) Additional displays shown in FIGS. **18** D and F may be used to indicate other types of enclosure or order issues. For example, the expediter may disable an enclosure as in the case of enclosure #4 as shown in FIG. **18**D by tapping or clicking on the “X” in the display. As shown in FIG. **18**F, the area **102** for enclosures #1 and #4 display information indicating that the orders in these enclosures liave been picked-up or collected. The information displayed in the area **102** for enclosure #1 indicates “Masha” order“#101” was “Collected”. The displays on the expediter monitor can also show other conditions encountered with orders which require follow-up. The displayed information shows two such conditions for the enclosures #15 and #9 shown in FIG. **18**F. The enclosure #15 display tells the expediter that “Patrick” has asked for help, a staff member will need to render assistance, and the expediter can clear the display by tapping on the menu button represented by the asterisks. The enclosure #9 display tells the expediter that there has been a food error in the “Mike” order delivered to the enclosure, that he should recall the order and food delivered to the enclosure and use the menu button represented by the three asterisks to reset the display.

(72) Each representation of the enclosures or “Boxes” #1 through #20 in the displays shown in FIGS. **18**A-F includes a menu button represented by three dots located adjacent the enclosure number. The expediter may tap or click on these menu buttons which will allow the expediter to perform various functions such as disabling a box or opening the door for a customer.

(73) A summary of activity in the various enclosures or “Boxes” **100** may be provided on the screen of the expediter monitor **42**. That summary may be displayed in areas **114** as shown in shown in FIGS. **18** D-F. With respect to the activity as shown in FIG. **18**E, the following summary is displayed: “1 order”. With respect to activity as shown in FIG. **18**D, the following summary is displayed: “5 orders” and“1 expired”. Although not included in the area **114** shown in FIG. **18**D, the display could include “1 disabled”. With respect to the activity as shown in FIG. **18**F, the following summary is displayed in the area **114**: “1 request”, “1 error”, “2 collected”, “7 orders”, “2 expired”, “1 disabled”.

(74) In order to handle issues arising with customers as they try to collect their food from the enclosures, the displays on the expediter screen as shown in FIGS. **18**A-F may be made available to a customer service person. This may be accomplished by providing an APP which may be downloaded

to a tablet or other device available for use by the customer service person, where the APP provides the same displays and functionality to the tablet or other device being used 80y the customer service person.
(75) The display on the monitor **28** as shown in FIG. **19** displays information to customers concerning “ACTIVE ORDERS” or “ORDERS IN PROCESS”. Unlike the display in FIG. **7**, the display to customers as shown in FIG. **19** describes the status of some orders as only “RECEIVED” and the status of other orders as “COOKING” so as to provide the customer with more information about the progress of their order.
(76) FIGS. **20-22** show three different, dynamic displays which may appear on the monitor **26** to advise customers that their orders are ready to be collected from one or more assigned enclosures or “BOXES” **18**. In FIG. **20**, there are only three “READY TO COLLECT” orders so the space dedicated to each order on the screen of the monitor is large thereby maximizing the size of each customer's name and the number of the enclosure. As shown in FIG. **21**, the display must accommodate twenty customer orders which are ready to collect so the space available for and the size of the customer's name and assigned enclosure are necessarily limited. However, in order to assist a customer with recognizing that their order is ready to collect, the display shown in FIG. **22** appears on the monitor **26** for a few seconds as soon as the order is ready (i.e., “CLAIRE” in enclosure **9**) and then reverts to the display shown in FIG. **21** with the customer's name being added to the list in FIG. **21**.
(77) FIG. **23** illustrates a series of steps **1-9** representing the interaction between touch screens of the kitchen order preparation monitor **46** and control order control monitor **48** (i.e., the “expediter monitor”) shown in FIGS. **9**, **10** and **18**(A-F) which are under the control of the order computer **45** and the wall computer **44** shown in FIG. **3**. As shown in FIG. **23**, these computers **45** and **44** as well as the digital memory storing the necessary programmable instructions are depicted by block **116**. Steps **1** and **2** as shown are directed to the use of interactive displays on the touch screen of the kitchen order preparation monitor **46** by the kitchen order preparer to make the orders presented and then signaling when each order has been prepared and ready for the expediter to deliver each order. Steps **3-10** as shown are directed to the interactive displays of FIGS. **18**A-F which are read, used and/or created by the expediter.
(78) While the invention is susceptible to various modifications and alternative constructions, only certain illustrative embodiments have been shown and described in the drawings and accompanying detailed description. It should be understood, however, that there is no intention to limit the invention to the specific construction and embodiments disclosed herein. On the contrary, the invention is intended to cover all modifications, alternative constructions, embodiments, and equivalents falling within the scope of the invention as claimed herein.

Claims

1. A method for dispensing customer food orders using an array of food enclosures for receiving customer food orders from a food supply area, each of the enclosures including an opening having a movable door closing the opening, and a food order and array management computer system comprising at least one digital processor and a digital memory associated with the digital processor, the digital memory storing programmed instructions for performing the following steps: receiving customer food orders associated with customer personal devices; electronically displaying in the food supply area availability of enclosures so as to allow customer food orders to be supplied to empty enclosures and unavailability of enclosures which contain previously supplied customer food orders, electronically displaying in the food supply area the assignment and supplying of each of the customer food orders to the enclosures; electronically displaying customer-specific order information in the food supply area for all of the customer food orders while the orders are contained in the enclosures; and receiving transmissions from the respective customer personal devices enabling the customers to open the movable door on the respective enclosures to which the respective customer food orders have been supplied.

2. The method of claim 1 further comprising electronically displaying in the food supply area customer-specific order information including a customer name associated with each of the customer food orders while the orders are contained within the enclosures.
3. The method of claim 1 further comprising electronically displaying in the food supply area customer-specific order information including an order number associated with each of the customer food orders while the orders are contained within the enclosures.
4. The method of claim 1 further comprising electronically displaying in the food supply area customer-specific order information including the time of delivering each of the customer food orders while the orders are contained within the enclosures.
5. The method of claim 1 further comprising electronically displaying in the food supply area customer-specific order information including a pick-up time specified by a customer for the customer food order while the orders are contained within the enclosures.
6. The method of claim 4 further comprising electronically displaying in the food supply area customer-specific order information including an alert that the difference between the pick-up time specified by the customer and the actual time has exceeded a predetermined limit.
7. The method of claim 1 further comprising electronically displaying in the food supply area a queue of customer food orders to be delivered to the enclosures.
8. The method of claim 1 wherein each movable door of each of the enclosures remains locked after customer food orders are supplied from the food supply area and is subsequently unlocked by customers using the respective personal devices, the method further comprising the following steps: transmitting a door opening instruction to the respective personal device of each customer specifying the necessary response to unlock the door of an enclosure containing at least a portion of the customer food order; and receiving the necessary response from the personal device of the customer to unlock the door.
9. The method of claim 8 wherein the door opening instruction is transmitted by SMS.
10. The method of claim 9 wherein the necessary response is transmitted by SMS.
11. The method of claim 9 wherein the necessary response is transmitted over the Internet.
12. The method of claim 11 wherein the necessary response transmitted to a customer-specific URL.
13. The method of claim 1 wherein the customer food orders are received from the customer personal devices.